EXPERIMENT 2: Entropy, Cross Entropy, and Perplexity

AIM:

To compute Entropy, Cross Entropy, and Perplexity for two text corpora. These information-theoretic measures are fundamental to evaluating language models.

PROCEDURE:

1. Import numpy as np.
2. Define two text samples: text_A and text_B.
3. Define a preprocess() function to lowercase, remove non-alphabetic characters, and split into tokens.
4. Tokenize both texts.
5. Build vocabulary from each text.
6. Compute unigram probabilities for each vocabulary.
7. Calculate Entropy: H(P) = -sum(p * log2(p)).
8. Calculate Cross Entropy: H(P, Q) = -sum(p(w) * log2(q(w))).
9. Calculate Perplexity: PP = 2^(Cross Entropy).
10. Print Tokens A, Tokens B, Entropy of each corpus, Cross Entropy, and Perplexity.

CODE:

```
import numpy as np
import re

text_A = "Natural language processing enables computers to understand human language"
text_B = "Language models learn patterns in text and understand language"

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.split()

tokens_A = preprocess(text_A)
tokens_B = preprocess(text_B)

vocab_A = set(tokens_A)
vocab_B = set(tokens_B)

# Unigram probabilities
def unigram_prob(tokens):
    probs = {}
    total = len(tokens)
    for word in tokens:
        probs[word] = probs.get(word, 0) + 1
    for word in probs:
        probs[word] /= total
    return probs
```

```
prob_A = unigram_prob(tokens_A)
prob_B = unigram_prob(tokens_B)

# Entropy
entropy_A = -sum(p * np.log2(p) for p in prob_A.values())
entropy_B = -sum(p * np.log2(p) for p in prob_B.values())

# Cross Entropy H(A||B)
cross_AB = -sum(prob_A.get(w, 1e-10) * np.log2(prob_B.get(w, 1e-10))
                for w in vocab_A)

# Perplexity
perp_AB = 2 ** cross_AB

print("Tokens A:", tokens_A)
print("Tokens B:", tokens_B)
print("\nEntropy of Corpus A:", entropy_A)
print("Entropy of Corpus B:", entropy_B)
print("\nCross Entropy (A || B):", cross_AB)
print("\nPerplexity:", perp_AB)
```

OUTPUT:

```
Tokens A: ['natural', 'language', 'processing', 'enables', 'computers', 'to',
'understand', 'human', 'language']
Tokens B: ['language', 'models', 'learn', 'patterns', 'in', 'text', 'and',
'understand', 'language']

Entropy of Corpus A: 2.94770277922009
Entropy of Corpus B: 2.94770277922009

Cross Entropy (A || B): 22.9806067441743

Perplexity: 8276599.654616931
```

RESULT:

Entropy, Cross Entropy, and Perplexity were successfully computed for the two corpora. Both corpora have equal entropy (~2.95 bits), indicating similar word distribution uniformity. The high cross entropy (22.98) and perplexity (~8.27 million) indicate that corpus B is a poor model for predicting corpus A, since they share limited vocabulary overlap.